CHAPTER B

Fluency bank

Two questions per session, in 6 to 8 minutes including the correction. The identifiers allow revisits; they are not a list to be done in full at every lesson. The availability dates are based on the planned teaching, and should be shifted if the topic has not yet been studied.

R001 – Fractions

Calculate $\frac{2}{3} + \frac{1}{4}$.

R002 – Order of operations

Calculate $5 - 2(3 - 7)$.

R003 – Powers

Simplify $\dfrac{10^5 \times 10^{-3}}{10^4}$.

R004 – Roots

Simplify $\sqrt{50}$.

R005 – Proportion

18 represents 30% of a total. What is the total?

R006 – Direct rate

A price of 80 falls by 15%. Final value?

R007 – Initial value

After a 20% increase, the value is 72. Initial value?

R008 – Calculated rate

A quantity goes from 40 to 50. Rate?

R009 – Successive rates

A 20% increase then a 20% decrease: overall rate?

R010 – Reciprocal rate

Which rate compensates a 20% decrease?

R011 – Zero product

Solve $(2x-1)(x+3)=0$.

R012 – Expansion

Expand $(2x-3)^2$.

R013 – Factorisation

Factorise $3x^2-12x$.

R014 – Sign of an affine function

Solve $-3x+6\geq 0$.

R015 – Slope

Slope of the line through $(1,4)$ and $(3,0)$?

R016 – Line

Equation of the line of slope -2 through $(1,4)$?

R017 – Weighted mean

The values 2 and 5 have counts 3 and 2. Mean?

R018 – Median

Median of 1, 2, 2, 5, 8, 10?

R019 – Quartiles

For the data set 1, 2, 3, 4, 5, 6, 7, 8, give Q_1 and Q_3 with the ceiling ranks $\left\lceil \frac{n}{4} \right\rceil$ and $\left\lceil \frac{3n}{4} \right\rceil$.

R020 – Standard deviation of data

For 1 and 3 with equal weights, mean, variance and standard deviation?

R021 – Units

Convert 1 min 40 s to seconds; 2400 frames over this duration give how many per second?

R022 – Simple probability

With a fair die, probability of a multiple of 3?

R023 – Complement

$P(A) = \frac{3}{8}$. Give $P(\overline{A})$.

R024 – Interval

Write $x \geq -2$ and $x < 3$ as an interval.

R025 – Logic

Negate “all the students passed.

R026 – Histogram

Two classes of widths 2 and 4 have the same count 20. In a histogram with areas proportional to counts, what are the relative heights?

R027 – Pie chart

A sector measures $72°$. Proportion?

R028 – Scale

An axis starts at 90. Bars stop at 100 and 110. Is the second value double the first?

R029 – Percentage of a percentage

40% of the students take A; among them, 25% take B. What proportion takes A and B?

R030 – Proportion and count

A category represents 0.125 of 80 people. Count?

R031 – Defining and representing a sequence

Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$; suggest an explicit formula.

R032 – Arithmetic sequences

We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine r and u_0.

R033 – Geometric sequences

Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

R034 – Sums, indices and proofs

Calculate $\sum_{k=2}^{5} 2^k$ without the summation symbol in the answer.

R035 – Conjecturing a limit and programming a threshold

For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

R036 – Three forms, three readings

Give three forms of $g(x) = x^2 + 2x - 3$. Read off its roots and its minimum.

R037 – Discriminant and solving

Solve $3x^2 - 6x + 3 = 0$ and $x^2 + 2x + 5 = 0$.

R038 – Factorising without automating the discriminant

Determine the polynomial f of degree 2 such that $f(1) = f(4) = 0$ and $f(0) = -8$.

R039 – Sign, inequalities and optimisation

Solve $x^2 - 4x - 5 < 0$, then maximise $x(8 - x)$ on $[0, 3]$.

R040 – Projection and angle

Two vectors have norms 2 and 5 and form 90°, then 180°. Calculate the products.

R041 – Coordinates, norm and orthogonality

For which real number t are the vectors $(t, 2)$ and $(3, -6)$ orthogonal?

R042 – Vector calculus and identities

Given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

R043 – Al-Kashi and choosing a method

A triangle has $AB = AC = 4$ and $BC = 4\sqrt{2}$. Determine the angle A.

R044 – The circle with diameter AB

For $A(0,0)$, $B(6,0)$, determine the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.

R045 – Rate of change and secants

Calculate the rate of $f(x) = x^2 + 3x$ between 2 and $2 + h$.

R046 – Derivative at a point

For $f(x) = 3x^2 - 2x$, calculate $f\prime(1)$ from the rate.

R047 – Tangent and affine approximation

Given $f(3) = 5$ and $f\prime(3) = -2$. Write the tangent and approximate $f(3.04)$.

R048 – When differentiability fails

Justify that $g(x) = |x - 2|$ is not differentiable at 2. Also study the differentiability of $x \mapsto \sqrt{x}$ at 0 through its rate.

R049 – Conditioning: changing the denominator

Out of 60 objects, 20 are red; among the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

R050 – Trees, partitions and total probability

$P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

R051 – Independence and disjointness

Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.18$. Are they independent?

R052 – Successive trials and Bernoulli trials

Four independent trials, each with probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

R053 – Simulating and discussing the model

A simulation of 2000 points gives 675 points below $y = x^2$. What area does it estimate?

R054 – Derivative function and standard functions

Differentiate x^5, x^{-2} and $\sqrt{x}$, stating the domains.

R055 – Sum, product, reciprocal and quotient

Differentiate $(x^2 + 1)(3x - 2)$, then $\frac{1}{x^2 + 1}$.

R056 – Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on $[0, 3]$ and give its extrema on this interval.

R057 – Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

R058 – Optimisation and Newton's method

A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation.

R059 – Normal vector and Cartesian equation

Equation of the line through $(2, -1)$ with normal $(3, 4)$.

R060 – Orthogonal projection and distance

Project $P(4, 1)$ onto $d : x = 2$.

R061 – Equation of a circle

Recognise $x^2 + y^2 + 2x - 8y + 17 = 0$.

R062 – Solving a configuration

Intersect $(x - 1)^2 + (y + 2)^2 = 9$ with $y = 1$.

R063 – Variable, event and distribution

Two fair coins are tossed. X counts the heads. Write its distribution and $P(X \leq 1)$.

R064 – Expectation and linearity

A gross gain is 10 with probability $\frac{1}{5}$ and 0 otherwise. What stake makes the game fair?

R065 – Variance, standard deviation and König–Huygens

For X with values $-1, 1$ equally likely, calculate the expectation, variance and standard deviation.

R066 – Transforming and comparing gains

$E(X) = 3$, $V(X) = 4$. Give the characteristics of $Y = -2X + 5$.

R067 – Samples and observed mean

For $\mu = 5$, $\sigma = 3$ and n=36, give the experimental interval asked for.

R068 – Defining a new function

What is the equation of the tangent to e^x at 1?

R069 – Algebraic properties and comparisons

Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

R070 – Differentiating and studying

Differentiate $g(t) = 3e^{-2t}$ and study its direction of variation.

R071 – From discrete to continuous

For $Q(t) = 5e^{0.3t}$, express $Q(n)$ and $\frac{Q(t+2)}{Q(t)}$.

R072 – Building numerically and proving an inequality

Show that $e^{2x} \geq 1 + 2x$ for every real number x.

R073 – Measuring an angle in radians

Convert 45°, 120°; give the arc for radius 2 m and angle $\frac{3\pi}{4}$.

R074 – Wrapping and the image of a real number

Place the images of $\frac{7\pi}{2}$ and $-\frac{5\pi}{3}$ by reducing them to within one turn.

R075 – Sine, cosine and standard values

If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

R076 – Related angles and symmetries

Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

R077 – Approximating pi and putting it to use

For n=6 and c=1, calculate the new chord after doubling.

R078 – Reading a graph

On the figure R078, read $f(1)$, solve $f(x) = 0$ and $f(x) > 0$.

